Baroreceptor Regulation of Blood Pressure

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Adapted from: Baroreceptor Regulation of Blood Pressure

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe baroreceptors and their role in the regulation of blood pressure.
2. Diagram the major anatomic components of the baroreceptor reflex.
3. Explain how changes in arterial blood pressure affect the baroreceptor reflex.
4. Define autonomic tone and describe its relationship to the baroreceptor reflex.

CASE CONNECTION

After playing three rigorous sets of tennis, you decide to take it easy and sit down for a while. You talk to your partner for about 30 minutes and then stand up to walk to your car. Almost immediately you feel lightheaded. Within a few minutes, it goes away. On your way home, you open a bottle of water and drink all of it in 30 seconds. You do not experience any additional lightheadedness.

What caused your lightheadedness when you stood up? What compensatory mechanisms came into play? Consider your answers as you read, and we'll revisit the questions at the end of the brick.

GO TO CONCLUSION ↓

Introduction

The regulation of the cardiovascular system is to ensure uninterrupted oxygen delivery to tissues during a variety of activities. This is accomplished by rapid and slower systems: the rapid autonomic nervous system (ANS); and the slower circulating factors like vasopressin, natriuretic peptides, and angiotensin II, and local factors like nitric oxide that respond in specific tissue beds. The ANS response can alter heart rate, cardiac contractility, and systemic vascular resistance, and this means the autonomic response needs to be carefully regulated.

What Is the Role of Baroreceptors in Blood Pressure Regulation (LO1)?

The body has sensors, **baroreceptors,** in key locations so it can detect decreases or increases in blood pressure (BP) or blood volume. The body can then homeostatically correct these changes by the appropriate neural, or hormonal mechanisms. The **baroreceptor system** is one such regulator.

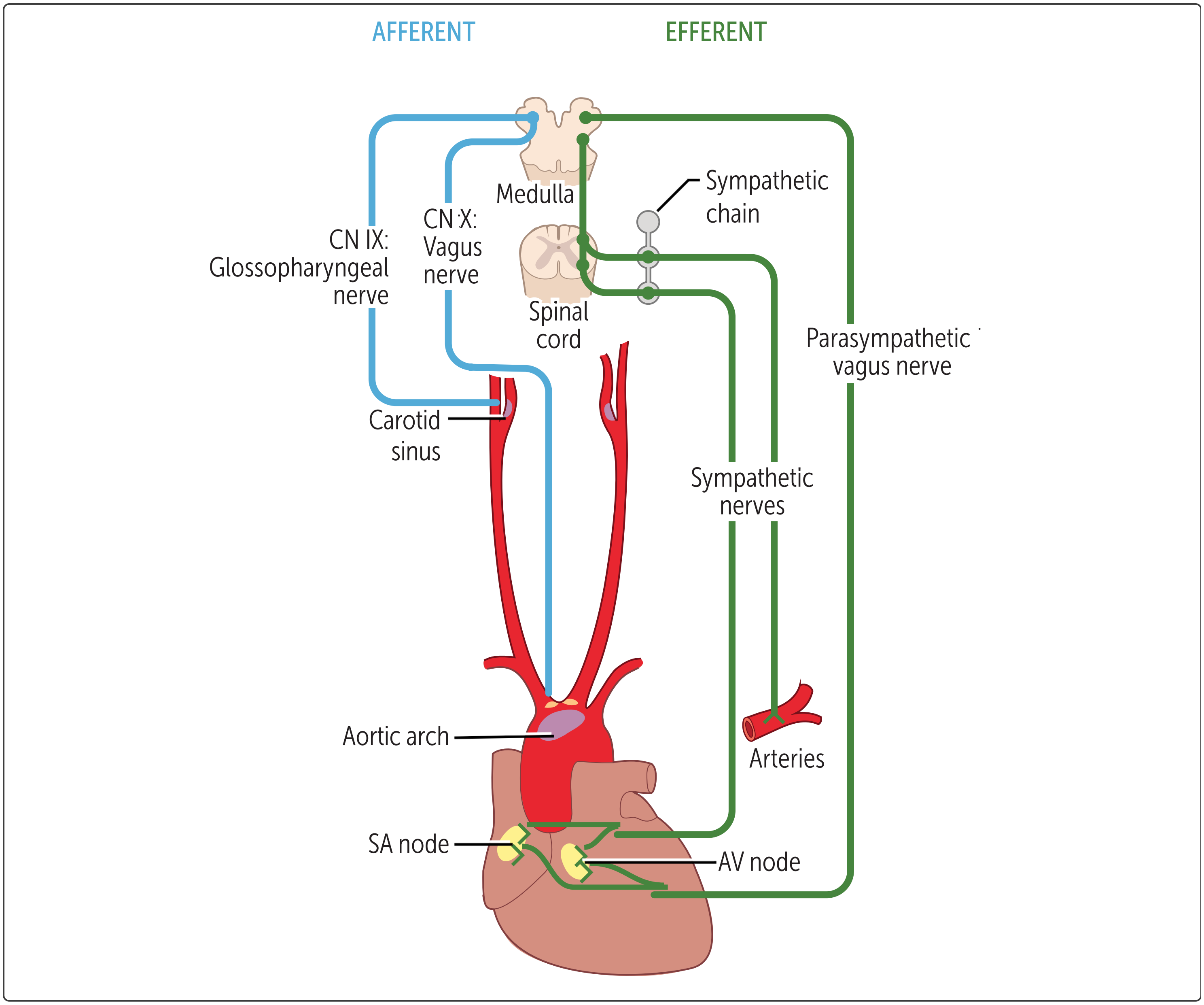
Baroreceptors are specialized pressure sensors that detect **stretch within blood vessel** walls and then stimulate a response from the nervous system. This stretch detection mechanism is crucial for maintaining appropriate blood pressure throughout the body. When blood volume increases within a vessel, pressure rises, causing the vessel walls to stretch. These stretched baroreceptors then activate and transmit signals to the medulla in the brainstem. Then signals are sent to effectors in the body that will regulate BP by altering vessel diameter, heart rate, and cardiac contractility.

Why do we need to regulate blood pressure? As you can probably imagine, there must be some sort of limit to how much blood a vessel can hold; too much blood in one location could overstretch and damage the vessel. However, too little blood could deprive your organs of oxygen. This is where baroreceptors come in: they moderate swings in blood pressure through the **baroreceptor reflex**. This integrated response helps maintain blood pressure within safe physiological limits, ensuring adequate perfusion to all tissues while protecting vessel integrity.

Baroreceptors sense the stretch within a blood vessel and then send signals to the brain, which subsequently coordinates the appropriate response.

What Is the Anatomy of the Baroreceptor Reflex (LO2)?

Baroreceptors are located throughout the body's arterial vasculature, though the greatest density of receptors lies close to the heart; specifically, at the **aortic arch and carotid sinus** (Figure 1). Because the reflex results in altered cardiac function, it makes sense that these sensors would be close to the heart. Their role is to detect changes in blood pressure via changes in arterial vessel stretch. More pressure equals more stretch.



QUIZ

Tap image for quiz

Figure 1 Anatomy of baroreceptor reflex

When the baroreceptors in these locations are activated by changes in stretch, a signal is sent to the medulla (an afferent signal) for processing (see blue lines in Figure 1). A subsequent output signal is sent from the medulla (an efferent signal) to the heart's sinoatrial (SA) and atrioventricular (AV) nodes (see green lines in Figure 1), the cardiac myocytes (not shown), and blood vessels. These afferent and efferent signals make up the neural circuitry required for the baroreceptor reflex. Let's now detail which nerves make up these afferent and efferent components.

Afferent signals to the brain are carried by:

- Carotid sinus afferent nerves: cranial nerve IX (glossopharyngeal)
- Aortic arch afferent nerves: cranial nerve X (vagus nerve)

Efferent autonomic signals (think - E for exit) from the brain are carried by:

- Sympathetic nerves (traveling via the spinal cord)
- Parasympathetic nerves (traveling in cranial nerve X [also called the vagus nerve])

Again, refer to Figure 1 to place these nerves into the context of the baroreceptor reflex.

Cranial nerves IX (glossopharyngeal) and X (the vagus nerve) are involved in the baroreceptor reflex.

How Does the Baroreceptor Reflex Work (LO3)?

The **baroreceptor reflex** is the body's way of regulating acute, beat-to-beat changes in blood pressure (BP). If BP increases or decreases even slightly, the baroreceptor reflex aims to bring it back to normal (to achieve homeostasis). This reflex is the reason BP does not typically fluctuate much from moment to moment. As with all homeostatic responses, the baroreceptor reflex is something that happens automatically. Let's now detail the sensor and then the actions that result from the reflex.

The Sensor: Arterial Stretch

The **arterial baroreceptors** fire on each heartbeat, but the frequency of firing increases when there is a greater arterial stretch (ie, higher BP), and the frequency decreases when the BP falls (Figure 2).

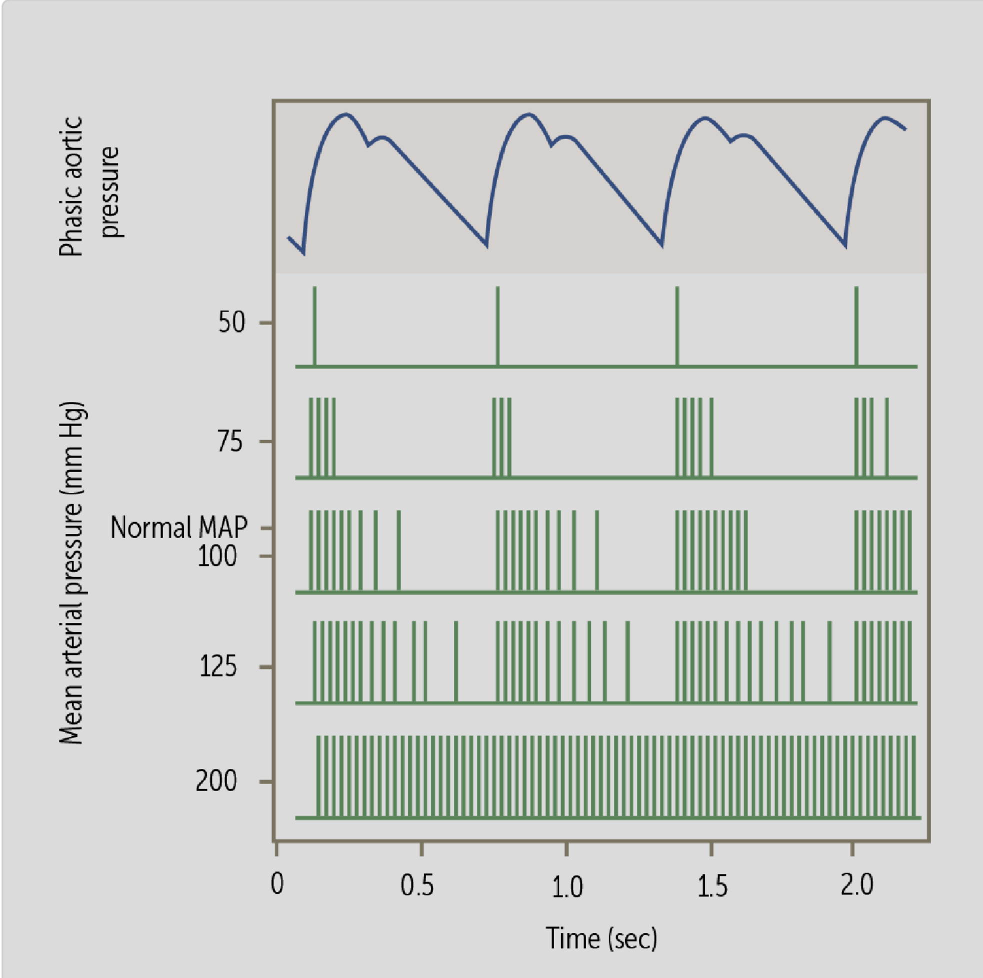


Figure 2 Firing of aortic arch baroreceptor at different blood pressures

The baroreceptor is sensitive to a wide range of BPs and can respond to both increases and decreases in BP on a beat-to-beat basis. Figure 2 demonstrates the firing of baroreceptors at different mean arterial blood pressures (MAP):

- At normal MAP, each time the heart contracts (as shown in the phasic aortic pressure wave at the top), the baroreceptors respond by firing a series of signals.
- As blood pressure decreases (shown in the top two rows of green lines below the aortic pressure tracing), the firing frequency of baroreceptors diminishes significantly with each heartbeat.

- Conversely, when blood pressure increases (shown in the bottom two rows with more densely packed vertical green lines), the baroreceptor firing rate per heartbeat increases substantially, reaching maximum frequency at a MAP of about 200 mm Hg.
- Note: a MAP of 200 mm Hg represents a dangerously elevated blood pressure that could potentially cause serious complications like stroke or heart attack. These signals are then transmitted by cranial nerves IX and X to the brain for processing.

An increase in arterial stretch, as seen with higher BP, would increase the firing rate of baroreceptors. Conversely, a decrease in arterial stretch, as seen with lower BP, would decrease the firing rate of baroreceptors.

Afferent Signal to the medulla

The baroreceptor signals that sense **high blood pressure (BP)** are sent to the medulla, from which efferent signals are sent to the body in response. These signals are sent via efferent nerves to the following structures:

- **SA and AV nodes of the heart:** the heart rate drops, reducing cardiac output (CO) and BP.

- **Cardiac myocytes:** their contractility (force of contraction- mainly the ventricles) drops, reducing cardiac output and BP.
- **Arteries:** The smooth muscles relax, causing arterial dilation and a decrease in systemic vascular resistance (SVR), which reduces blood pressure (BP). Note SVR can also be called total peripheral resistance (TPR).
- **Veins:** the smooth muscle relaxes (venodilation), which lowers venous pressure and increases capacitance, thus decreasing venous return to the heart. This decreases preload (ventricular stretch) and lowers cardiac stroke volume (SV).

All these actions will lower the BP back toward normal. The exact opposite sequence would happen if the blood pressure fell, which would reduce the arterial stretch. Then, the efferent signals would prompt a faster heart rate, greater cardiac contractility, venoconstriction, and an increase in systemic vascular resistance via vasoconstriction.

Which specific efferent outputs from the medulla cause the overall response described above? For high BP, there is suppression of the sympathetic nervous system (SNS), and this reduce SNS activity slows HR, reduces contractility, and decreases sympathetic arteriolar tone and increases venodilation. On the flipside, there is stimulation of the parasympathetic nervous system, which causes slowing of the heart rate. The parasympathetic nervous system has minimal effect on contractility and the blood vessels. The opposite happens in hypotension, with the above summarized in Table 1. Note that the baroreceptor afferent firing increases with increased arterial pressure, whereas medullary efferent output changes in the opposite direction to restore blood pressure toward normal.

Table 1 Baroreceptor responses

Stimulus	Barorecept. effect	Barorecptor afferent firing	Medulla efferent firing	SNS end organ effect	PSNS end organ effect
↑ Blood pressure	↑ Stretch	↑	↓ SNS ↑ PSNS	↓ HR ↓ Contractility ↓ Vascular resistance Venodilation	↓ HR (SA/AV nodes)
↓ Blood pressure	↓ Stretch	↓	↑ SNS ↓ PSNS	↑ HR ↑ Contractility ↑ Vascular resistance Venoconstriction	↑ HR (SA/AV nodes)

Increased arterial pressure activates the baroreceptors, leading to **inhibition of sympathetic activity** (reducing heart rate, inotropy, and vasoconstriction) and **enhancement of parasympathetic activity** (decreasing cardiac output via reduced heart rate). Combined effects lead to a decrease in blood pressure.

All would increase. Losing the afferent fibers would lead to a diminished rate of baroreceptor activity. Lower baroreceptor activity means there is less inhibition of the SNS, so the heart rate, vasoconstriction, and contractility would increase.

Let’s look at the mechanism of baroreceptor reflex in response to **low blood pressure** (Figure 3).

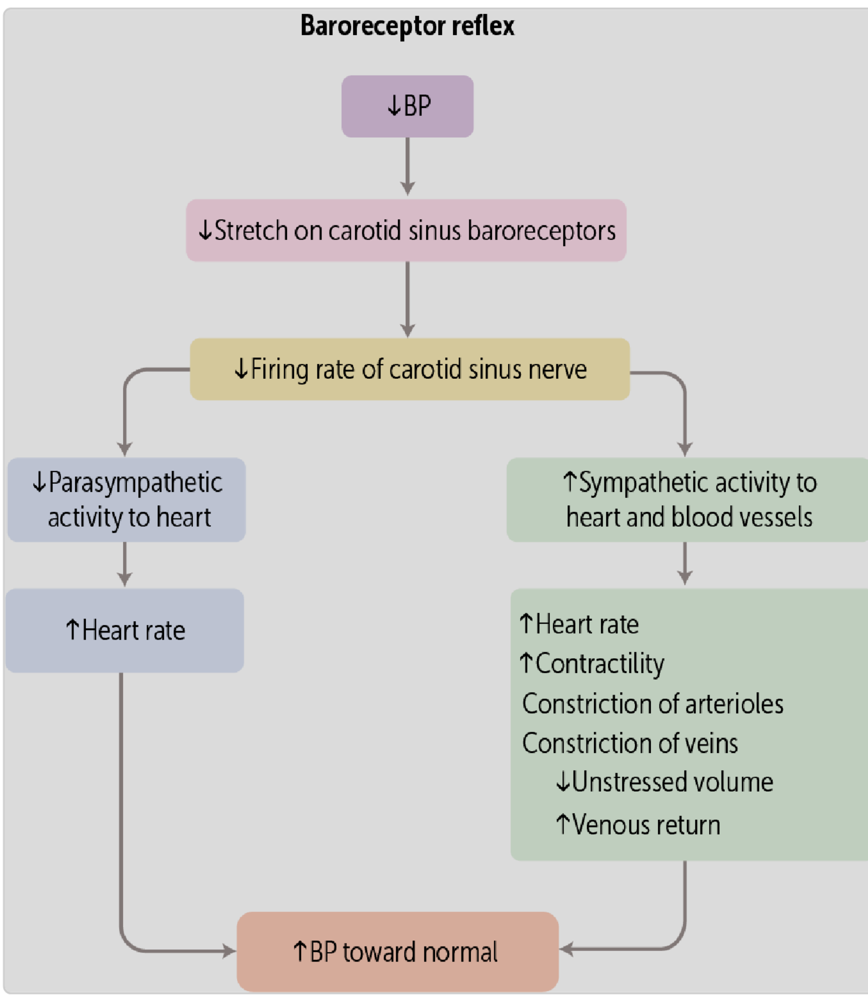


Figure 3 Mechanism of baroreceptor reflex

Baroreceptor reflex on the venous side of the heart: If the baroreceptor reflex is a response to increased stretch on the arterial side, you may wonder if there is an **analogous autonomic response on the venous (right) side of the heart**. Yes, it's called the **Bainbridge reflex**, but it is not very important in humans. It causes tachycardia (by inhibiting parasympathetic outflow) when the right atrium is stretched. The right atrium typically stretches when there is volume overload and high venous return to the heart. The tachycardia relieves the volume overload by increasing the cardiac output.

Baroreceptors in the carotid sinus and aortic arch detect **arterial pressure** because it directly influences perfusion to vital organs, while venous pressure fluctuations are less critical for immediate autonomic regulation (Bainbridge reflex).

What Is Autonomic Tone (LO4)?

While discussing autonomic outputs during the baroreceptor reflex, it is worth mentioning an important concept related to neural transmission. Autonomic nerves, such as those involved in the baroreceptor reflex, constantly send out information (See Fig 2). They have a constant baseline rate of firing, even in the absence of overt stimulation by the afferent signals. This is known as the **tonic activity** of a nerve. Implicit in this is that vasodilation can occur above and below this baseline, where changes in vessel resistance affect blood pressure and flow.

Why is autonomic tone important for the baroreflex? The sympathetic and parasympathetic fibers that carry the efferent responses from the medulla to the heart and blood vessels do not lie dormant between signals from the baroreceptors. They have a constant baseline activity of firing at normal blood pressures. Thus, the firing rate can increase or decrease from baseline, allowing for optimal control when blood pressure changes. This makes sense when you think about the name "autonomic" nervous system. It implies a steady activity, even without outside stimuli, and that is, indeed, a characteristic of these nerves.

Autonomic tonicity is the constant baseline firing rate of autonomic nerves that occurs even when they are not receiving afferent signals.

CASE CONNECTION BACK TO INTRODUCTION

Thinking back to your lightheadedness after tennis, what caused it? What compensatory mechanisms came into play?

As a result of the fluid loss with exercise (sweating) and venous pooling when you stood up, your preload was reduced. Your blood pressure decreased, and lightheadedness followed because the blood flow to your brain decreased. Compensatory mechanisms then came into play. Baroreceptors in the aortic arch and carotid sinus sensed the decrease in preload. The sympathetic nervous system then sent signals to increase heart rate, myocardial contractility, and vasoconstriction. When you rehydrated yourself by drinking fluids, the reverse set of physiologic changes occurred.

Summary

- Baroreceptors are sensors in key locations of the body that detect changes in blood pressure (BP) and volume and provide that information to the brain.
- High-pressure baroreceptors detect acute changes in blood pressure by monitoring stretches in the arterial vasculature.
- The high-pressure baroreceptors are primarily found in the aortic arch and carotid sinus.
- Baroreceptors send afferent signals to the medulla.
- The baroreceptor reflex is the body's way of regulating acute, beat-to-beat changes in BP; if the BP starts to get too high or too low, the baroreflex aims to bring it back to normal.
- In the baroreceptor reflex, an increase in blood pressure leads to greater stretching of the arteries, which, in turn, causes reductions in heart rate,

cardiac contractility, and peripheral vascular resistance; the reverse occurs with a decrease in blood pressure.
- If the baroreceptor senses hypertension, the sympathetic output decreases and the parasympathetic output increases; the reverse occurs for hypotension.
- Autonomic nerves display tonicity, meaning they have steady baseline firing activity.
- That firing rate can increase or decrease from baseline, allowing for optimal control when the blood pressure changes.

Review Questions

1. The carotid sinus baroreceptor is most sensitive to detecting changes in which of the following?

   A. Arterial pressure

   B. Cardiac contractility

   C. Heart rate

   D. Left ventricular preload

   E. Velocity of blood flow

Hide Explanation

Correct answer A). Recall that baroreceptors respond to changes in arterial pressure by detecting arterial stretch. The nerve endings associated with baroreceptors lie within the tunica adventitia (outermost layer of a blood vessel wall) of arteries. As more blood fills the vessel, the pressure within the lumen increases, the adventitia is stretched, and the baroreceptor reflex is triggered. Incorrect answers B). Cardiac contractility and C). heart rate applies most directly to the heart itself. D). Left ventricular preload refers to the volume that fills the heart during diastole. E). Velocity of blood flow refers to how fast or slow blood moves within the vasculature. Although increases in contractility, heart rate, and preload might indirectly increase the blood pressure and trigger the baroreceptor reflex, the blood pressure is the most direct stimulus.

2. A 26-year-old female presents to the emergency department with complaints of a "racing" heart rate. On electrocardiogram and exam, you determine she has supraventricular tachycardia, an abnormally fast heart rhythm (an arrhythmia). One technique for the treatment of arrhythmias is the "carotid massage." This is done by applying firm pressure on the neck overlying the carotid sinus. Which of the following best explains how this method treats some arrhythmias, particularly those with fast heart rates?

A. It directly stimulates the vagus nerve within the carotid sheath, slowing the heart rate.

B. It decreases the baroreceptor reflex, leading to decreased parasympathetic nervous system activity, thus slowing the heart rate.

C. It stimulates the baroreceptor reflex, leading to decreased sympathetic nervous system activity and thus slowing the heart rate.

D. It stimulates the baroreceptor reflex, leading to increased sympathetic nervous system activity, thus slowing the heart rate.

Hide Explanation

Correct answer D). Performing a carotid massage applies direct pressure to the carotid sinus, which stimulates the arterial baroreceptors at the area and increases their firing rate. The higher firing rate will lead to the stimulation of the parasympathetic nervous system and the inhibition of the sympathetic nervous system (SNS), both of which will cause a decrease in heart rate. It is this decrease in heart rate that ultimately helps treat some types of arrhythmias. Incorrect answers: A). Compressing the nerve does not slow the heart rate. B). Carotid massage stimulates the baroreceptor reflex, and does not decrease it, and D). An increase in SNS activity is the opposite of what would be seen.

3. A 23-year-old student goes wandering in the desert without his water bottle and develops severe dehydration. His blood pressure is 95/70 mm Hg and heart rate of 120/min. Which of the following would be ***least*** likely to occur as a result of the baroreceptor reflex response to this patient's hypotension?

A. Increase in cardiac contractility

B. Increase in cardiac output

C. Increase in sympathetic nervous system activity

D. Rapid pulse

E. Vasodilation of vasculature

Hide Explanation

Correct answer E). Dehydration decreases the effective circulating blood volume and would not be effective in this condition of dehydration, with the concomitant decreased blood volume, and hypotension. Also, the vascular smooth muscle will contract, not relax. Incorrect answers- for this question: This would manifest as a net decrease in blood pressure, which would then be detected through the baroreceptor reflex. This would cause the C). sympathetic efferent nerve to fire at a faster rate, leading to A). increases in cardiac contractility, and D). heart rate. Together, these will increase the B). cardiac output, the product of heart rate × stroke volume.

4. What would happen to heart rate, contractility, and vascular resistance if the afferent fibers of cranial nerves IX and X were cut?

A. Heart rate, contractility, and vascular resistance would all decrease.

B. Heart rate and contractility would increase, but vascular resistance would remain unchanged.

C. ○ Heart rate, contractility, and vascular resistance would all increase.

D. Only heart rate would increase, while contractility and vascular resistance would decrease.

E. No significant changes would occur.

Hide Explanation

Correct Answer: C. The loss of baroreceptor input due to severing cranial nerves IX and X would eliminate inhibitory feedback to the sympathetic nervous system, leading to an unchecked increase in heart rate, myocardial contractility, and systemic vascular resistance.

5. Which of the following describes the function of autonomic tone in the context of baroreceptor reflex?

A. Facilitates rapid adjustement of blood pressure by the ANS

B. Ensures perfusion pressure to essential organs is unaltered

C. Facilitates vasodilation regardless of baroreceptor signals

D. Maintains baseline vascular resistance at all times

E. Prevents changes in heart rate during postural transitions

Hide Explanation

Correct answer A). Changes in basal baroreceptor firing allow for rapid and automatic adjustments in blood pressure. Incorrect answers. B). Adequate perfusion pressure is affected by vessel tone, but it is the baroreceptor feedback that ensures constant perfusion pressure regardless of changes. C). Vascular tone responds to baroreceptor signals to maintain blood pressure, dilating and constricting as needed. D). Baseline vascular resistance is the normal state, but it is not the regulated homeostatic variable; blood pressure is. Thus, the vascular resistance is dynamically adjusted in response to the baroreceptor reflex. E). Again, the heart rate is not a homeostatically regulated variable, it changes to maintain blood pressure during postural changes.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe baroreceptors and their role in the regulation of blood pressure.~~

02 ☑ ~~Diagram the major anatomic components of the baroreceptor reflex.~~

03 ☑ ~~Explain how changes in arterial blood pressure affect the baroreceptor reflex.~~

04 ☑ ~~Define autonomic tone and describe its relationship to the baroreceptor reflex.~~

Objective Confidence: 4/4